

DHARSHAN CS

Backend Software Development Engineer proficient in Python, Flask, REST API design, PostgreSQL, and Redis.

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PROFESSIONAL SUMMARY

Backend Engineer skilled in building systems under real constraints. **Redesigned architecture** four times to scale a browser automation platform to **150 concurrent accounts**. Engineered a production Flask backend supporting **300-500 concurrent users** with sub-200ms response times. Proficient in Python, Flask, PostgreSQL, Redis, and Celery. Self-taught with dual degrees from IIT Madras and SLIET, demonstrating a track record of delivering end-to-end solutions from schema design to deployment.

EDUCATIONS

Indian Institute of Technology Madras | Chennai, India

Bachelor of Science in Data Science and Applications

Apr 2023 - Present

CGPA : 7.36

Sant Longowal Institute of Engineering and Technology | Punjab, India

Bachelor of Engineering in Mechanical Engineering

Nov 2022 - May 2026

CGPA : 7.40

PROJECTS

Text-to-Learn | AI Course Generator

Backend / Full-Stack Developer

Mar 2026 - May 2026

- Developed a full-stack platform that converts user prompts into structured courses, including modules, lessons, MCQs, and code blocks, utilizing a multi-stage **Gemini API** prompt pipeline.
 - Engineered a hierarchical **MongoDB** schema (Course → Module → Lesson) with **Mongoose**, creating 10+ RESTful endpoints for efficient CRUD operations across the content hierarchy.
 - Created a dynamic lesson renderer in **React** that supports five content block types, including syntax-highlighted code, embedded video queries, and interactive MCQs with answer explanations.
 - Protected user-specific routes using **Auth0 OAuth2**, facilitating persistent course storage and individual progress tracking across sessions.
- Technologies / Tools Used** : React, Node.js, Express, MongoDB, Gemini API, Auth0

ParkSpace v2 |  Github

Backend / Full-Stack Developer

Oct 2025 - Present

- Architected RBAC-enforced parking platform** with 3-tier permission hierarchies, sustaining 300–500 concurrent users under simulated production load.
 - Engineered 15+ REST API endpoints** with resource-based routing and structured error handling, serving a **Vue.js frontend at sub-200ms** average response time.
 - Automated async processing of CSV exports, scheduled emails, and background reports via **Celery + Redis**, **cutting task latency by 65%** and eliminating blocking ops.
 - Cut endpoint **response time from ~850ms to ~210ms via Redis caching** on PostgreSQL dashboard queries, reducing repeated database hits by 40%.
 - Modeled a **normalized 6-table relational schema** with referential integrity and dynamic pricing rules, sustaining zero data inconsistency under concurrent load.
- Technologies / Tools Used** : Python, Flask, PostgreSQL, Redis, Celery, Vue.js

Chegg Automation System

Developer

Mar 2024 - Dec 2024

- Engineered a **browser automation system** to eliminate manual 10-15 second refresh cycles, integrating **Telegram API** for real-time question alerts and completely removing screen monitoring.
 - Iterated through 4 architectural versions, scaling from 1 to 20+ concurrent accounts, diagnosing and resolving memory constraints caused by persistent browser contexts at each threshold.
 - Redesigned core architecture around a **cookie rotation model**—storing session-per-account to disk, utilizing a single event loop to cycle across all accounts and effectively scaling to 150 concurrent accounts on a single **EC2 instance** with no memory blowup.
 - Implemented automatic session recovery, including dead cookie detection, re-login, and fresh cookie injection without human intervention, achieving uninterrupted 24/7 operation.
- Technologies / Tools Used** : Python, Selenium, Pyppeteer, AWS EC2, Telegram API

SKILLS

Concepts :	REST API Design, RBAC, Async Task Processing, Session Management, Cookie Rotation, Load Testing (Locust)
Databases :	PostgreSQL, SQLite, Redis, MongoDB
Frameworks Libraries :	Flask, Vue.js, Bootstrap, Jinja2, SQLAlchemy, Selenium, Pyppeteer, Celery
Programming Languages :	Python, C++, Java, JavaScript, C, SQL
Tools Platforms :	Git, Linux, AWS EC2, Telegram API

AWARDS & ACHIEVEMENTS

- Leetcode Rated 1857** — Top 5% globally across **500,000+** active contestants
- Ranked 245th** out of **30,000+** participants in LeetCode Weekly Contest 497